

LECTURE 10

PANEL DATA MODELS (2)

- (A) Fixed Effects Estimator
- (B) Two-way Error Component Models
- (C) Random Effects Model

(A) Fixed Effects (FE) Estimator

- Consider a multiple-period panel data model:

$$y_{it} = \beta_0 + \beta_1 x_{it,1} + \cdots + \beta_K x_{it,K} + a_i + v_{it}, i = 1, \dots, n; t = 1, 2, \dots, T. \quad (1)$$

a_i : unobservable heterogeneity (fixed effect), correlated with the endogenous variable $x_{it,1}$.

- OLS estimators are biased and inconsistent. But we can remove the a_i by first differencing the data and then run the regression model as we did in usual.
- Some individuals (i.e., entities or cross-sectional units) may have different observations. For example, $T = 5$, but the individual 1 has two observations, and individual 2 has four observations only. This is the case of an unbalanced panel.

- For the unbalanced panel (and the balanced panel), we can use another way to remove a_i prior to estimation. It is called fixed effects transformation (or within transformation). For each i , we average the equation over time and obtain

$$\bar{y}_i = \beta_0 + \beta_1 \bar{x}_{i,1} + \cdots + \beta_k \bar{x}_{i,k} + \bar{a}_i + \bar{v}_i, \quad (2)$$

where $\bar{y}_i = \frac{1}{T} \sum_t y_{it}$, $\bar{x}_{i,1} = \frac{1}{T} \sum_t x_{it,1}$ and $\bar{v}_i = \frac{1}{T} \sum_t v_{it}$. Here, $\bar{a}_i = a_i$

- We subtract (2) from (1) for each t , and get

$$y_{it} - \bar{y}_i = \beta_1 (x_{it,1} - \bar{x}_{i,1}) + \cdots + \beta_k (x_{it,k} - \bar{x}_{i,k}) + (v_{it} - \bar{v}_i) \quad (3)$$

or

$$\ddot{y}_{it} = \beta_1 \ddot{x}_{it,1} + \cdots + \beta_k \ddot{x}_{it,k} + \ddot{v}_{it}, i = 1, \dots, n; t = 1, 2, \dots, T.$$

$\ddot{y}_{it} = y_{it} - \bar{y}_i$ is the **time-demeaned** data on y , and similarly for $\ddot{x}_{it,1}$ and $\ddot{v}_{it}$.

- The OLS estimators using the time-demeaned data (3) is called the **fixed effects estimator** or the **within estimator**, $\hat{\beta}_{1,FE}, \dots \hat{\beta}_{k,FE}$.
- If the explanatory variables are strictly exogenous, that is, the idiosyncratic error v_{it} is uncorrelated with explanatory variables across all the time periods, then the fixed effects estimators are unbiased and consistent.
- If the error v_{it} is only uncorrelated with the explanatory variables at the same time period t , the fixed effects estimators are consistent.

Example: We study the effect of alcohol taxes on traffic deaths in the U.S by using the fixed effect estimation. (Dataset: Panel_beertax.dta)

FE estimation removes the state effects a_i .

```
. xtreg vfrall beertax, fe
```

Fixed-effects (within) regression
Group variable: state

Number of obs = 336
Number of groups = 48

R-sq: within = 0.0407
between = 0.1101
overall = 0.0934

Obs per group: min = 7
avg = 7.0
max = 7

corr(u_i, Xb) = -0.6885

F(1,287) = 12.19
Prob > F = 0.0006

vfrall	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
beertax	-.6558736	.18785	-3.49	0.001	-1.025612	-.2861352
_cons	2.377075	.0969699	24.51	0.000	2.186212	2.567937
sigma_u	.7147146					
sigma_e	.18985942					
rho	.93408484	(fraction of variance due to u_i)				

F test that all u_i=0: F(47, 287) = 52.18 Prob > F = 0.0000

The result is qualitatively same as that of the first differenced estimation. That is, the effect of beer tax on traffic deaths is *negative* and significant.

Example: We study the return to education using the panel data set called WAGEPAN.dta.

$\ln(wage_{it})$

$$= \beta_0 + \beta_1 edu_{it} + \beta_2 black_{it} + \beta_3 hisp_{it} + \beta_4 exper_{it} + \beta_5 married_{it} + \beta_6 union_{it} + a_i + v_{it}, i = 1, \dots, n; t = 1, 2, \dots, T$$

Method 1: Pool all the cross-sections from different periods together.

Pooled OLS Estimation

```
. reg lwage educ black hisp exper expersq married union
```

Source	SS	df	MS	Number of obs = 4360		
Model	230.719766	7	32.9599665	F(7, 4352) = 142.61		
Residual	1005.80988	4352	.231114402	Prob > F = 0.0000		
				R-squared = 0.1866		
				Adj R-squared = 0.1853		
Total	1236.52964	4359	.283672779	Root MSE = .48074		

lwage	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
educ	.0993878	.0046776	21.25	0.000	.0902173	.1085583
black	-.1438417	.0235595	-6.11	0.000	-.1900303	-.0976531
hisp	.015698	.0208112	0.75	0.451	-.0251026	.0564985
exper	.0891791	.010111	8.82	0.000	.0693563	.1090019
expersq	-.0028487	.0007074	-4.03	0.000	-.0042354	-.0014619
married	.1076656	.0156965	6.86	0.000	.0768925	.1384387
union	.1800726	.0171205	10.52	0.000	.1465076	.2136375
_cons	-.0347057	.064569	-0.54	0.591	-.1612938	.0918824

Fixed Effects Estimation

>xtreg lwage educ black hisp exper expersq married union,fe

```
. xtreg lwage educ black hisp exper expersq married union ,fe
note: educ omitted because of collinearity
note: black omitted because of collinearity
note: hisp omitted because of collinearity
```

```
Fixed-effects (within) regression               Number of obs   =       4360
Group variable: nr                             Number of groups =        545

R-sq:  within = 0.1780                         Obs per group:  min =         8
          between = 0.0005                      avg =        8.0
          overall = 0.0638                      max =         8

corr(u_i, Xb)  = -0.1139                       F(4,3811)       =       206.38
                                          Prob > F        =       0.0000
```

lwage	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
educ	(omitted)					
black	(omitted)					
hisp	(omitted)					
exper	.1168467	.0084197	13.88	0.000	.1003392	.1333542
expersq	-.0043009	.0006053	-7.11	0.000	-.0054876	-.0031142
married	.0453033	.0183097	2.47	0.013	.0094056	.081201
union	.0820871	.0192907	4.26	0.000	.044266	.1199083
_cons	1.06488	.0266607	39.94	0.000	1.012609	1.11715
sigma_u	.4000539					
sigma_e	.35125535					
rho	.5646785	(fraction of variance due to u_i)				

```
F test that all u_i=0:      F(544, 3811) =      7.98      Prob > F = 0.0000
```

```
. estimates store fixed
```

- **Limitation of FE approach**

The fixed effects estimation (just like the first differenced estimation) cannot estimate the partial effects of time-invariant explanatory variables, e.g., gender, location etc., on the y variable.

Because, all the time-invariant variables are removed along with the fixed effect α_i when we use within transformation.

Should we use the fixed effects or the first differenced estimation?

- When $T = 2$, the FE and FD estimates and all test statistics are identical.
- When $T \geq 3$, the FE and FD estimators are not the same.

- Since FD & FE estimators are unbiased and consistent, for large n and small T , the choice between FE & FD hinges on the relative efficiency of the estimators.
 - If the errors v_{it} are serially uncorrelated (given that they are homoskedastic), the FE estimator is more efficient.
 - However, if the errors v_{it} follow a random walk, then FD estimator will be more appropriate as v_{it} 's first differenced will be serially uncorrelated.
- Always try both estimations, and if the results are not very sensitive. That's good.

Remarks:

- (a) Testing in FD and FE regressions is the same as that in OLS regression since FD and FE are OLS after the first-differenced or within transformation.
- (b) For the large n and small T (i.e., $n \gg T$) dataset, it is called micropanel data.
- (c) For the small n and large T (i.e., $n \ll T$) dataset, it is called macropanel data.

(B) Two-way Error Component Models

- Consider the panel data model:

$$y_{it} = \beta_0 + \beta_1 x_{it,1} + \cdots + \beta_k x_{it,k} + u_{it}, \quad u_{it} = \mu_i + v_{it}, \quad (A)$$

a_i : unobservable time-invariant fixed effect (heterogeneity), i.e., a_i is constant over time but differs across individuals (entities)

- The advantage of using panel data sets is to control unobservable heterogeneity. In the fixed effects estimation, we can not only control for a_i but also time effects that are constant across entities (i.e., all cross-sectional units have the same time effect at time t) but evolve over time.
- For example, the safety improvements in new cars are introduced nationally (i.e., the states are cross-sectional units) over the years and they serve to reduce traffic deaths. If we ignore (and omit) these improvement changes over the time and do regression analysis, we may have the omitted variable bias.

- Time effects only:

$$y_{it} = \beta_0 + \beta_1 x_{it,1} + \cdots + \beta_k x_{it,k} + f_t + v_{it}$$

f_t : unobservable (individual-invariant) variable, accounting for any *time-specific* effect that is not included in the explanatory variable, called time effect.

Examples:

- (a) Global shocks (oil crisis, financial crisis), year-specific variables (2008, 2009) affecting all individuals.
- (b) Government (automobile safety) campaigns in specific years
- (c) A wide spread optimism or pessimism in the year 2000 (or the year 1929) in the stock market affecting all individuals.

- The time effect f_t could be correlated with the explanatory variables. If we ignore and omit it, the regression model suffers the endogeneity problem.

- Generalization: composite error

$$u_{it} = a_i + f_t + v_{it} \quad (B)$$

The equation (B) is called the 2-**way error** component model; the equation (A) is call the **one-way error** component model.

Recall: $u_{it} = a_i + v_{it}$, (A)

Example: alcohol taxes and traffic deaths in the U.S. (Panel_beertax.dta)

$$vfrall_{it} = \beta_0 + \beta_1 beertax_{it} + a_i + f_t + v_{it}$$

a_i : state effects;

f_t : time (year) effect

We can control the time effects by including time dummies.

The fixed-effects regression: control for state effects & time effects

> xtreg vfrall beertax y82 y83 y84 y85 y86 y87, fe

```
xtreg vfrall beertax y82 y83 y84 y85 y86 y87, fe
```

```
fixed-effects (within) regression      Number of obs   =      336
group variable: state                  Number of groups =      48
```

```
!-sq:                                Obs per group:
    within = 0.0803                    min =      7
    between = 0.1101                   avg =     7.0
    overall = 0.0876                   max =      7
```

```
corr(u_i, Xb) = -0.6781                F(7,281)         =      3.50
                                          Prob > F         =     0.0013
```

vfrall	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
beertax	-.6399799	.1973768	-3.24	0.001	-1.028505	-.2514551
y82	.0518038	.0396235	1.31	0.192	-.0261927	.1298003
y83	-.0280991	.0397307	-0.71	0.480	-.1063066	.0501085
y84	-.0206167	.0395898	-0.52	0.603	-.098547	.0573135
y85	-.0721725	.0390447	-1.85	0.066	-.1490297	.0046848
y86	.0139393	.0387718	0.36	0.719	-.0623808	.0902595
y87	.0009018	.0384704	0.02	0.981	-.0748249	.0766285
_cons	2.376666	.0985106	24.13	0.000	2.182754	2.570579
sigma_u	.70945965					
sigma_e	.18788295					
rho	.93446372	(fraction of variance due to u_i)				

```
test that all u_i=0: F(47, 281) = 53.19                Prob > F = 0.0000
```

(C) Random Effects (RE) Model

- Consider the panel data model

$$y_{it} = \beta_0 + \beta_1 x_{it,1} + \cdots + \beta_k x_{it,k} + a_i + v_{it}, \quad i = 1, \dots, n; t = 1, 2, \dots, T \quad (4)$$

The unobserved heterogeneity a_i is the individual specific (and time-invariant) factor. If the a_i is correlated with explanatory variables, then we could fix the endogeneity problem by using either the FD or FE estimation.

- However, the FD and FE methods cannot estimate the partial effects of time-invariant explanatory variables, e.g., race, gender, location, etc., on the y variable.
- When the effects of these time-invariant explanatory variables are interesting and important, e.g., racial wage gap, gender wage gap, location premium, etc., we need to find a way to estimate them.

- An alternative to FD and FE models is the random effects (RE) model. To use the RE model, we have an important assumption.

➤ The unobserved heterogeneity a_i is uncorrelated with the explanatory variables (and v_{it})

$$Cov(x_{it,j}, a_i) = 0, t = 1, 2, \dots, T; j = 1, 2, \dots, k$$

- If the explanatory variables are exogenous, the OLS estimators are unbiased and consistent.

- Suppose the $u_{it} = a_i + v_{it}$. The equation becomes:

$$y_{it} = \beta_0 + \beta_1 x_{it,1} + \dots + \beta_k x_{it,k} + u_{it}. \quad (5)$$

Because a_i is constant over time, u_{it} are serially correlated (across time):

$$Cov(u_{it}, u_{i,t+1}) = Cov(a_i + v_{it}, a_i + v_{i,t+1}) = Var(a_i) > 0,$$

$$Var(u_{it}) = Var(a_i + v_{it}) = Var(a_i) + Var(v_{it}).$$

(Here a_i and v_{it} are uncorrelated)

- The consequence of OLS with serial correlated errors is that OLS is inefficient.
- To deal with the serial correlation “problem”, we define

$$\lambda = 1 - \sqrt{\frac{\sigma_v^2}{\sigma_v^2 + T\sigma_a^2}}$$

where $\sigma_a^2 = \text{Var}(a_i)$ and $\sigma_v^2 = \text{Var}(v_{it})$. $0 \leq \lambda \leq 1$.

Then, we transform the equation (5) to below,

$$y_{it} - \lambda \bar{y}_i = \beta_0(1 - \lambda) + \beta_1(x_{it,1} - \lambda \bar{x}_{i,1}) + \cdots + \beta_k(x_{it,k} - \lambda \bar{x}_{i,k}) + (u_{it} - \lambda \bar{u}_i) \quad (6)$$

where $\bar{y}_i, \bar{x}_{i,1}, \dots, \bar{x}_{i,k}$ denote the time averages.

- After the transformation, errors $(u_{it} - \lambda \bar{u}_i)$ are serially uncorrelated.

$$\text{Cov}(u_{it} - \lambda \bar{u}_i, u_{i,t+1} - \lambda \bar{u}_i) = 0. \quad (\text{Exercise})$$

The OLS for the equation (6) is called the **Random Effects (RE)** estimation.

- The RE estimator is similar to FD and FE estimators. FD and FE are the OLS of first-differenced equation (data) and fixed effects transformed equation (time-demeaned data), respectively.
- Equation (6) is called the **quasi-transformation** and $y_{it} - \lambda \bar{y}_i$ is called **quasi-demeaned data**.
- For $\lambda = 1$, Equation (6) = FE estimation. [Check the page 3, Equation (3)]
- For $\lambda = 0$, Equation (6) = no transformation. [Usual OLS]

Remarks:

(a) λ is unknown. To do the transformation we need to estimate λ first,

$$\lambda = 1 - \sqrt{\frac{\sigma_v^2}{\sigma_v^2 + T\sigma_a^2}}$$

where $\hat{\sigma}_a^2$ and $\hat{\sigma}_v^2$ are consistent estimators of σ_a^2 and σ_v^2 . Many econometrics packages, e.g., Stata, automatically computes some version of $\hat{\lambda}$.

(b) The quasi-transformation allows for time-invariant explanatory variables, and this is the advantage of using the RE estimation, e.g., we can include gender in the RE model.

(c) In many applications, the whole reason for using panel data is to allow the unobserved effect to be correlated with the explanatory variables.

(d) Then, should we use FE/FD or RE estimation?

Example: wage equation using panel data (WAGEPAN.dta)

Coefficients of (time-invariant) educ, black, hisp

Random-effects Estimation:

```
>xtreg lwage educ black hisp exper expersq married union, re
```

```
. xtreg lwage educ black hisp exper expersq married union, re
```

```
Random-effects GLS regression           Number of obs   =       4360
Group variable: nr                     Number of groups  =        545
```

```
R-sq:  within = 0.1774                  Obs per group: min =         8
       between = 0.1837                  avg           =        8.0
       overall = 0.1808                  max           =         8
```

```
Random effects u_i ~ Gaussian           Wald chi2(7)      =       943.95
corr(u_i, X)      = 0 (assumed)         Prob > chi2      =       0.0000
```

lwage	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
educ	.1012246	.0089133	11.36	0.000	.0837549	.1186943
black	-.1441307	.0476148	-3.03	0.002	-.237454	-.0508073
hisp	.0201511	.0426011	0.47	0.636	-.0633456	.1036477
exper	.1121195	.0082609	13.57	0.000	.0959285	.1283105
expersq	-.0040689	.0005918	-6.88	0.000	-.0052288	-.0029089
married	.0627951	.0167729	3.74	0.000	.0299209	.0956693
union	.1073789	.01783	6.02	0.000	.0724327	.142325
_cons	-.1074643	.1107057	-0.97	0.332	-.3244435	.1095149
sigma_u	.32456727					
sigma_e	.35125535					
rho	.46057172	(fraction of variance due to u_i)				